

Using Kernel Density Estimates to Investigate Multimodality

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SUMMARY

A technique for using kernel density estimates to investigate the number of modes in a population is described and discussed. The amount of smoothing is chosen automatically in a natural way.

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1. INTRODUCTION

INVESTIGATION of the number of modes or maxima in a density or its derivative has been considered by several authors, for example Cox (1966) and Good and Gaskins (1980). Most methods seem to depend on some arbitrary implicit or explicit choice of the scale of the effects being studied; see the remarks of Silverman (1980). The simple approach based on kernel density estimates described in this note has the virtue of making this choice in an automatic and natural way. The test statistic used is defined in Section 2 below, and in Section 3 a technique for assessing significance is described. Finally in Section 4 an illustrative application is given.

2. THE CRITICAL WINDOW WIDTH

A possible test statistic for hypotheses concerning the number of modes in the density can be obtained by constructing kernel density estimates of the data. The kernel density estimate (Rosenblatt, 1956) for window width h based on univariate observations $X_1, \dots, X_n$ is defined by

$$\hat{f}(t; h) = n^{-1} h^{-1} \sum_{i=1}^n K\{h^{-1}(t - X_i)\}, \quad (1)$$

where K is a kernel function, which we shall assume throughout to be the normal density function. Apart from the theoretical advantages of this choice, the use of a normal kernel has strong computational advantages; see Silverman (1981).

The window width h controls the amount by which the data are smoothed to obtain the kernel estimate. Thus, for example, if the data are strongly bimodal a large value of h will be needed to obtain a unimodal estimate. Suppose that we wish to test the null hypothesis that the density f underlying the data has k modes, against the alternative that f has more than k modes; often $k = 1$. Define the k -critical window width h_{crit} by

$$h_{\text{crit}} = \inf\{h; \hat{f}(\cdot, h) \text{ has at most } k \text{ modes}\}. \quad (2)$$

Large values of h_{crit} will reject the null hypothesis. Silverman (1978) used a critical value of a smoothing parameter in a somewhat different context. The computation of h_{crit} in practice is facilitated by the following theorem and corollary.

Theorem. Given any fixed $X_1, \dots, X_n$, define $\hat{f}(t; h)$ as in (1) above, using a normal kernel K . For each integer $m \geq 0$, the number of maxima as t varies in $\partial^m \hat{f} / \partial t^m$ is a right continuous decreasing function of h .

The following corollary follows at once.

Corollary. Defining h_{crit} as in (2) above, $\hat{f}(\cdot, h)$ has more than k modes if and only if $h < h_{\text{crit}}$.

The corollary shows that h_{crit} can be found by a binary search procedure, since for any value of h we can tell at once whether or not $h < h_{\text{crit}}$ by counting the number of modes in $\hat{f}(\cdot; h)$. The result is also used in Section 3 below.

This section is concluded with the proof of the theorem, which makes use of the theory of total positivity; see, for example, Karlin (1968). Let $v_{m+1}(h)$ denote the number of sign changes in $\hat{f}^{(m+1)}(\cdot, h)$. Since $(-t)^{m+1} \hat{f}^{(m+1)}(t, h)$ is, for all $m \geq 0$ and h , eventually positive as $t \rightarrow -\infty$ and as $t \rightarrow \infty$, it suffices to show that v_{m+1} is decreasing and right continuous. For $h_2 > h_1 > 0$, $\hat{f}^{(m+1)}(\cdot, h_2)$ is the convolution of $\hat{f}^{(m+1)}(\cdot, h_1)$ with a $N(0, h_2^2 - h_1^2)$ density, and $\hat{f}^{(m+1)}(\cdot, h_1)$ is continuous and bounded. Thus, by Theorem 2 of Schoenberg (1950), $v_{m+1}(h_2) \leq v_{m+1}(h_1)$ so that v_{m+1} is decreasing. Now suppose, for given $h_0 > 0$, there exist $a_1 < b_1 < a_2 < \dots < a_r < b_r$ such that $\hat{f}^{(m+1)}(a_i, h_0) > 0$ and $\hat{f}^{(m+1)}(b_i, h_0) < 0$ for all i . By the continuity of $\hat{f}^{(m+1)}(t, \cdot)$, for all sufficiently small ε and all i , $\hat{f}^{(m+1)}(a_i, h_0 + \varepsilon) > 0$ and $\hat{f}^{(m+1)}(b_i, h_0 + \varepsilon) < 0$. Hence $\liminf_{h \downarrow h_0} v_{m+1}(h) \geq v_{m+1}(h_0)$; the right continuity of v_{m+1} follows from the fact that v_{m+1} is known to be decreasing.

Note that Schoenberg's theorem does not apply for general kernels. Indeed, the convolution of unimodal densities need not be unimodal; see Feller (1966, p. 164).

3. ASSESSING SIGNIFICANCE

For any particular k -modal simple null hypothesis, it is easy to assess, by simulation, the significance of the value of the critical window width obtained from the data. Suppose the null hypothesis is that the true density is g and that the value of h_{crit} obtained from the data is h_0 . Then the theory of Section 2 implies that

$$\text{pr}_g(h_{\text{crit}} > h_0) = \text{pr}\{\hat{f}(\cdot; h_0) \text{ has more than } k \text{ modes} \mid \{X_1, \dots, X_n\} \text{ is drawn from } g\}.$$

Thus, in order to assess the significance of h_0 for sample size n , it is only necessary to calculate the single density estimate $\hat{f}(\cdot; h_0)$ for each sample of size n generated from g ; there is no need to find h_{crit} for each replication.

The hypothesis that the true density is at most k -modal is of course a compound hypothesis. To provide a conservative assessment of the significance of h_0 , an appealing choice of the representative g_0 from which to simulate is obtained by rescaling $\hat{f}(\cdot, h_0)$, as constructed from the data, to have variance equal to the sample variance. The theory of Section 2 shows that g_0 is indeed at most k -modal; it is, in a sense, the most extreme k -modal density consistent with the data. It is extremely easy to simulate from g_0 ; Efron (1979) pointed out that independent observations y_i from g_0 are given by

$$y_i = (1 + h_0^2/\sigma^2)^{-1/2}(X_{I(i)} + h_0 \varepsilon_i),$$

where $X_{I(i)}$ are sampled uniformly, with replacement, from the data $X_1, \dots, X_n$, σ^2 is the sample variance of the data, and ε_i is an independent sequence of standard normal random variables.

Simulating from g_0 to assess significance is an example of a smoothed bootstrap procedure as defined by Efron (1979). However, Efron's procedure contains an implicit arbitrary choice of smoothing parameter, since his σ_z^2 is essentially arbitrary. In our case, the amount of smoothing is automatically determined in a natural way.

Finally, it should be pointed out that the theory and procedure of finding a critical window width and simulating from a rescaled density estimate constructed using this window width carries over immediately, *mutatis mutandis*, to the investigation of maxima in the first or higher derivative of the data. Both Cox (1966) and Good and Gaskins (1980) show a preference for seeking maxima in the density derivative.

4. AN APPLICATION

We illustrate the method by analysing a small data set of observations on chondrite meteors. These data consist of 22 observations which are given in Table 2 of Good and Gaskins (1980).

TABLE 1
*Chondrite data: critical window widths and their
estimated significance levels*

<i>Number of modes</i>	<i>Critical window width</i>	<i>P</i>
1	2.39	0.92
2	1.83	0.95
3	0.68	0.21
4	0.47	0.07

The data have been considered by several authors; see Good and Gaskins (1980) for details. In this analysis the raw values of the observations were used. Table 1 gives critical window widths and significance levels for tests of the null hypothesis that the underlying density has at most k modes against the alternative that it has more than k modes. The p -values are computed by simulating from a critical density as described above; 100 replications of 22 observations were used in each case.

These results must of course be interpreted as a hierarchical set of significance tests. All other things being equal, considerations of parsimony perhaps suggest that we should test successively for an increasing number of modes until we find a number that is accepted. Particularly bearing in mind the small sample size, the results clearly indicate the trimodal nature of the population; Good and Gaskins (1980) also arrived at this conclusion.

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